

Fall 19 Final exam practice sheet

The problems collected here are given as practice for the Final exam. These are *not* meant to be comprehensive and in order to do well you should study as many problems as possible. For example, any exam or practice exam will have problems that can be done. Also, don't forget quiz problems!

1. Determine if the following statements are True or False. No Justification is required.

(a) If $\sum_{n=0}^{\infty} a_n = L$, then $\sum_{n=0}^{\infty} a_n^2 = L^2$.

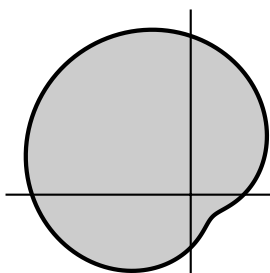
(b) If $\sum_{n=0}^{\infty} (-1)^n a_n$ diverges and all $a_n \geq 0$,
then $\sum_{n=0}^{\infty} a_n$ diverges.

(c) $\sum_{n=0}^{\infty} (f(n) - f(n+1)) = f(0)$.

2. Find $\int \frac{y^2}{(4-y^2)^{3/2}} dy$.

3. Find $\int x^a \ln(x) dx$ for $a > -1$.

4. Find the area inside the polar curve $r = 2 + \sin \theta - \cos \theta$ (shown below).



5. Find the volume when the region $0 \leq y \leq \sin^2(x)$ for $0 \leq x \leq \pi$ is spun about the y-axis.

6. Find $f^{(20)}(1)$ where $f(x) = \frac{x^2}{x^2 - 2x}$.

7. Set up, **but do not evaluate**, integrals which find the surface area when $y = \ln(x^2 + 1)$ for $0 \leq x \leq 1$ is spun around (a) the x-axis; and (b) the y-axis.

8. Find the volume from the solid resulting from the region $0 \leq y \leq \sec^3(x)$ for $0 \leq x \leq \frac{1}{4}\pi$ being spun around the x-axis

9. A storage tank has the shape of an upside down cone which is six feet across at the base and six feet high. The tank is currently full of a liquid with density 50 lbs/ft³ and is being emptied by being pumped out to a point 4 feet above the tip of the cone. Determine the total work done in emptying the tank.

10. For $a, b > 0$, show the following sum converges and find its value: $\sum_{n=0}^{\infty} \frac{(a^n + b^n)}{(a+b)^n}$. Simplify your answer for the value as much as possible.

11. Determine if the following sum converges or diverges. Justify your answer, including the name(s) of any test used.

$$\sum_{n=5}^{\infty} \left(\frac{2n^2 + e^{-n} + 17}{\ln(n) + 5n + 3n^2} \right)^n$$

12. Find the center and radius of convergence for the following series (you do not need to determine what happens at the endpoints).

$$\sum_{n=0}^{\infty} \frac{(3x-5)^n}{n^{24} 4^n}$$

13. Find the length of the parameterized curve $(\frac{1}{3}(t^2 + 6)^{3/2}, 3t)$ for $0 \leq t \leq 3$.

14. You are engaged in your favorite hobby of picking grapes and you are working along a particular steep hill. At the bottom of the hill your bag is empty and weighs 2 pounds; at the top of the hill you have gained in elevation 20 feet and your bag now weighs 17 pounds. If you make the assumption that the grapes were added in at a constant rate as you went up the hill, then determine the work done in moving the bags and grapes up the hill.

15. Find the value for $\int_0^{\pi} x^7 \cos(x^5) dx$ expressed as an (infinite) series.

16. Express $0.166166166166\dots$ as a geometric series. Find the sum of the series to write the number as a fraction.